

- Supervised learning technique
- Predicts continuous numerical values
- Models relationship between: Independent variables / features (X) and Dependent variable / target (y)
- Find the function that best maps inputs to outputs

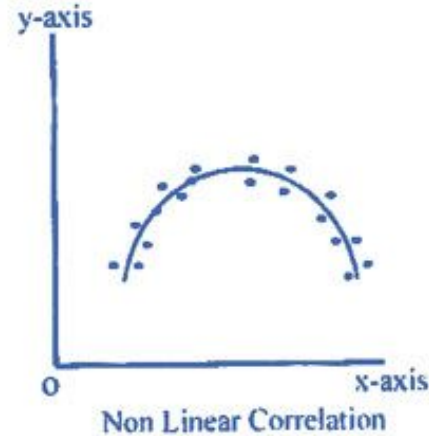
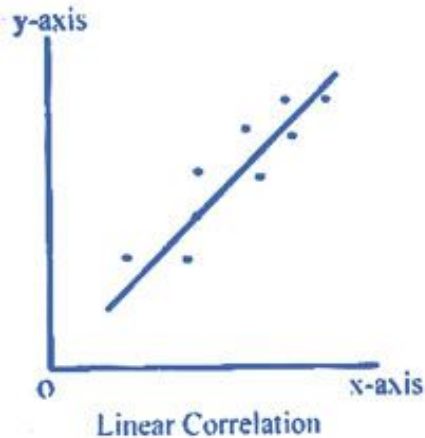
Examples

- House price prediction (Size, Bedrooms, Location)
- Salary prediction (Experience, Education)
- Sales forecasting (Ad Spend, Price, Season)
- Student Performance Prediction (StudyHours, Attendance)

- Linear Regression
 - Simple Linear Regression
 - Multiple Linear Regression
- Polynomial Regression
- ...

When to Use Linear Regression?

- Relationship between features and target is linear
- Target variable is continuous



- Model the relationship between a single feature (X) and the target (y) using a straight line. The goal is to find the line that best fits the data points.

$$y = mx + c$$

Where:

m = slope

c = intercept

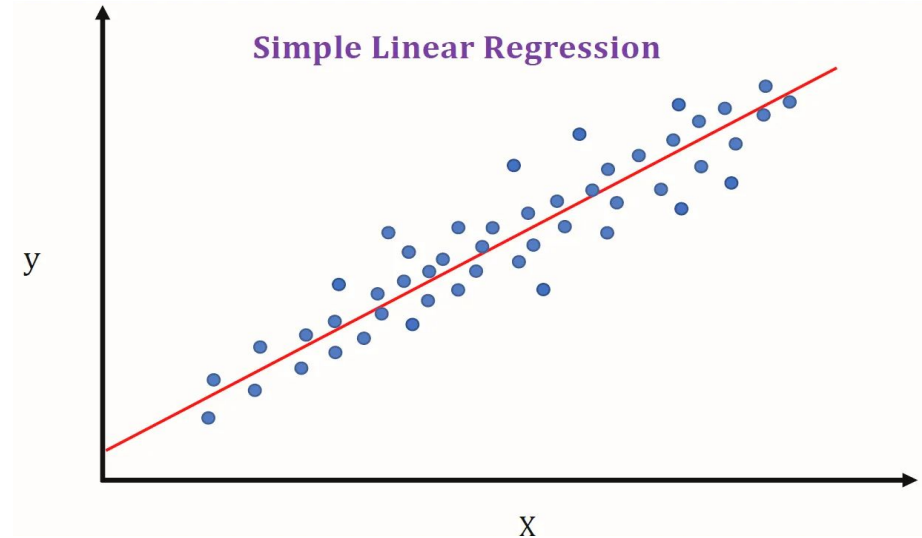
x = feature

y = predicted value

Note: m and c are model parameters

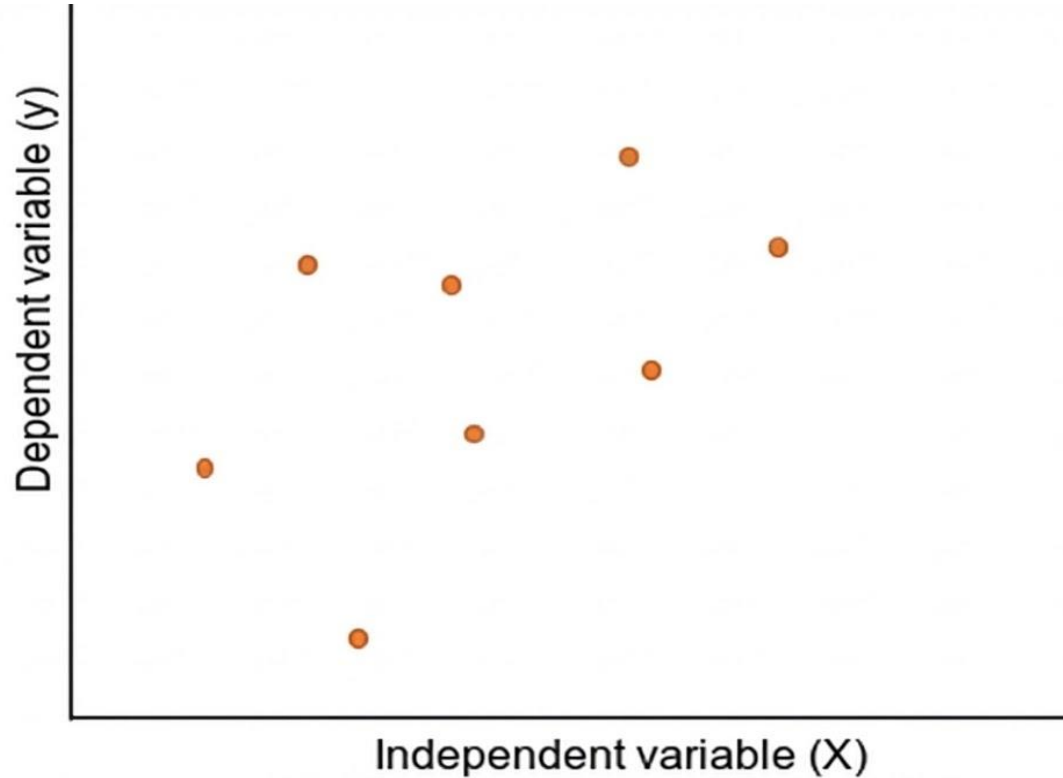
Or we can write as $y = wx + b$; w = weight, b = bias.

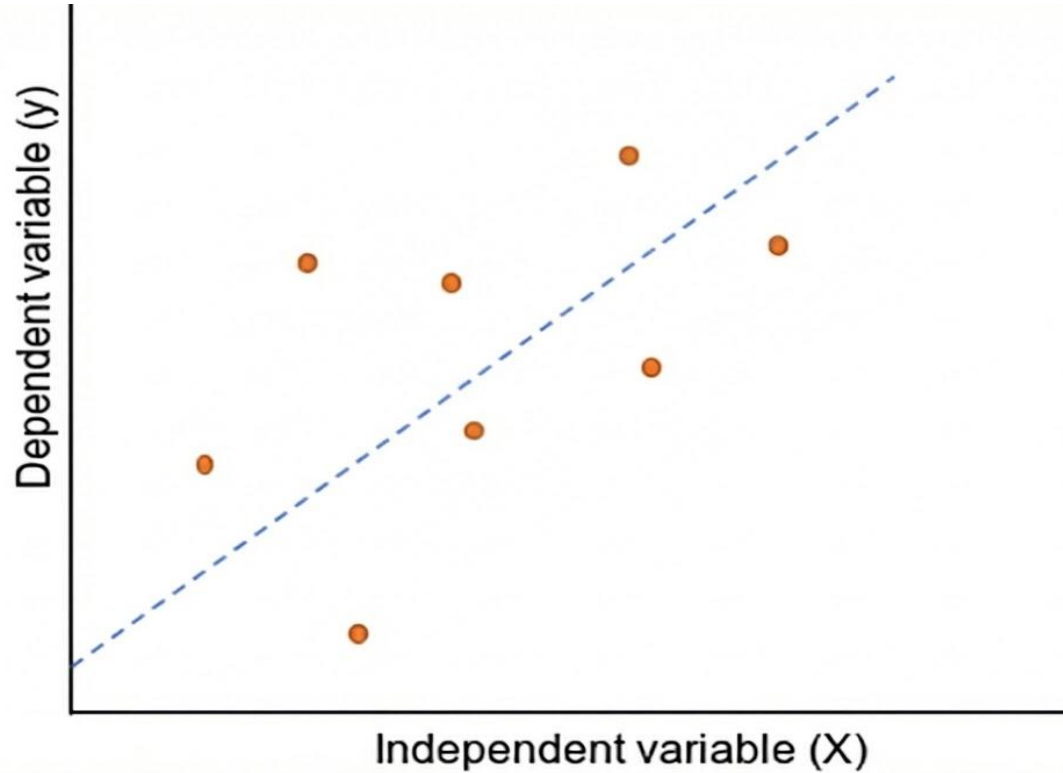
Or $y = w_1x + w_0$ Or $y = w_0x_0 + w_1x_1$ where $x_0=1$

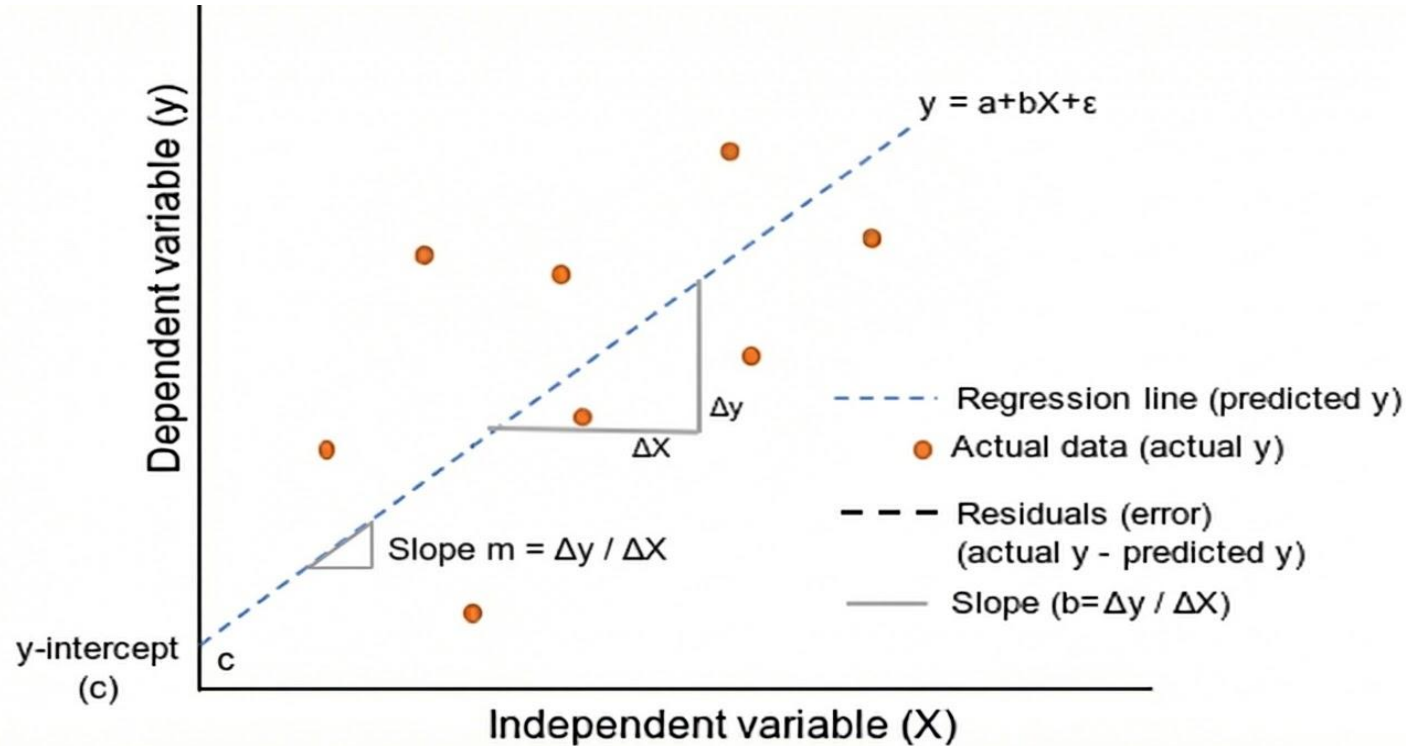


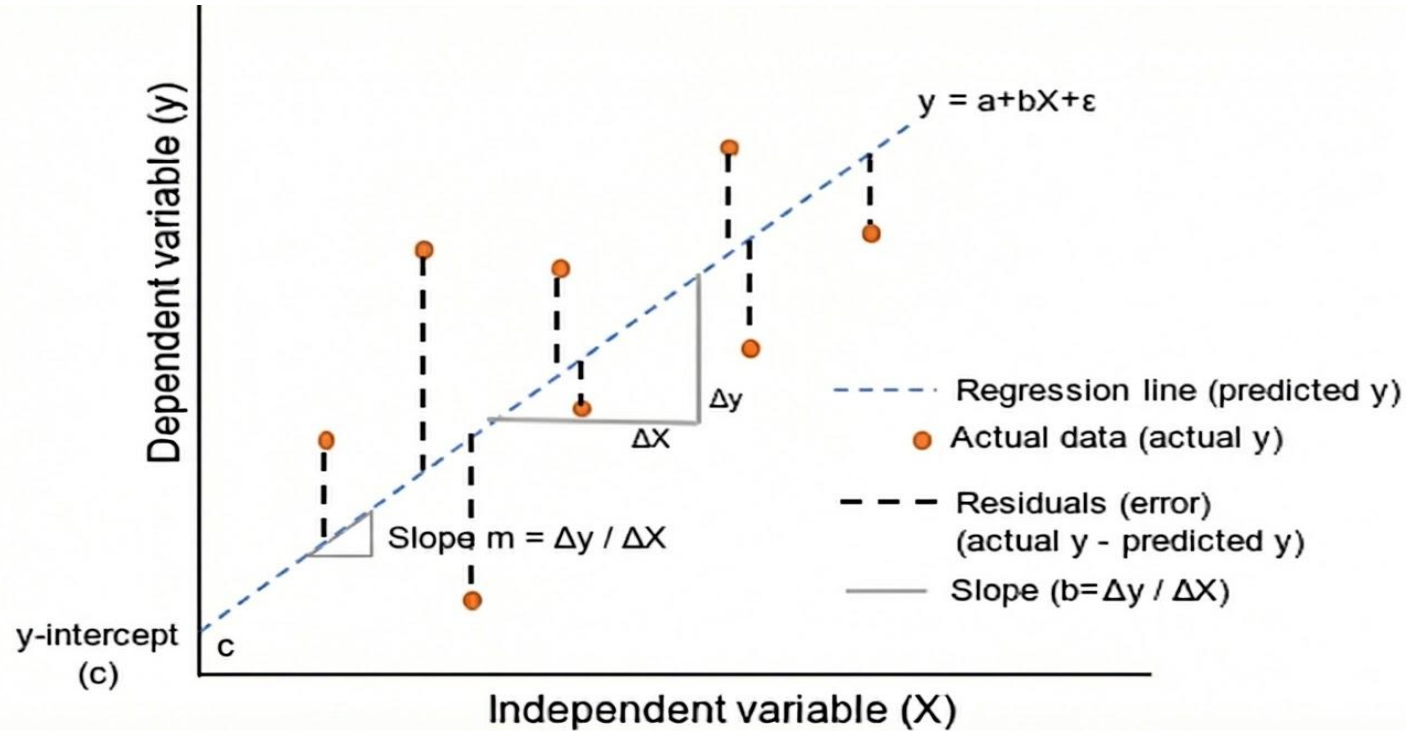
Simple Linear Regression

Visualize









Simple Linear Regression

What Makes a Line "Best"?

- Infinite lines can be drawn through data points. Which one is optimal?
- The best line is the one that minimizes the total error between actual and predicted values
- Residual (error) = Actual – Predicted.
- Loss: Measures error for one prediction. Calculated for a single data point. Absolute loss = $|y - \tilde{y}|$
- Cost: Measures error for overall predictions. Calculated for the entire dataset
- If we only sum errors, a very bad model can look perfect because mistakes cancel each other.
- **Cost function** determines optimal line.

- Mean Absolute Error (MAE) calculates the average of absolute differences between predicted and actual values. It treats all errors equally regardless of magnitude.
- $MAE = (1/n) \sum |actual - predicted|$
- Less sensitive to outliers.
- It penalizes errors linearly — a 10-unit error is exactly twice as bad as a 5-unit error.
- The absolute function is not differentiable at zero, making mathematical optimization slightly harder.

- MSE measures the average of the squares of the errors
- $MSE = (1/n) \sum (\text{actual} - \text{predicted})^2$
- Squaring removes negative signs (no error cancellation)
- Penalizes large errors more heavily. A 10-unit error becomes 100, 5-unit error becomes 25.
- Smooth and differentiable → easy optimization
- Sensitive to outliers

Minimize the Cost Function (MSE) by finding the optimal Slope (m) and Intercept (c).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - (mx_i + c))^2$$

To find the minimum of any function:

- Take derivative
- Set derivative = 0
- Solve

So we:

- Take derivative of MSE w.r.t. m
- Take derivative of MSE w.r.t. c
- Set both equal to 0
- Solve the two equations

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$c = \bar{y} - m\bar{x}$$

- Slope m measures how X and Y vary together
- Intercept c ensures line passes through center of data
- The best-fit line always passes through the mean point $(\bar{x}, \bar{y})$

When there are p features (independent variables), the model becomes:

$$y = w_0 + w_1x_1 + w_2x_2 + \dots + w_px_p$$

$$Y = XW$$

Matrices

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix} \quad W = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_p \end{bmatrix}$$

- First column = all 1s (for the bias w_0)
- Each row = one data point
- x_{ij} = value of the j -th feature for the i -th example

- Cost Function (MSE): $J(W) = \frac{1}{n} \sum (Y - XW)^2$
- Minimizing the Cost: Take derivative with respect to W , set to 0.

$$W = (X^T X)^{-1} X^T Y$$

During training:

- We minimize a cost function (usually MSE)
- Goal → Find optimal parameters (weights)

After training:

- We evaluate how good the model actually is
- Goal → Measure prediction performance

Training = Optimization

Evaluation = Performance Measurement

- MAE → Average absolute prediction error
- MSE → Average squared error
- RMSE → Square root of MSE (error in original units)
- R^2 Score → How well the model explains the data (compared to predicting the mean)

Interpretation

- If MAE = 5 → On average, predictions are off by 5 units
- Higher MSE → Larger errors → Worse model
- R^2 close to 1 → Good model
- R^2 close to 0 → Worse model. Same as predicting the mean

Lower MAE / MSE / RMSE is better. Higher R^2 is better.

- It shows how well the data fit a regression line or curve.
- It compares your model to a very simple baseline: predicting the average.
- MSE tells us how large errors are. R² tells us how good the model is relative to predicting the mean.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

sum of squared
residuals,

$$SS_{res} = \sum (y - \hat{y})^2$$

total sum of squares,

$$SS_{tot} = \sum (y - \bar{y})^2$$

